

EXPERIMENT 7 – Banker's Algorithm

Aim: To implement Deadlock Avoidance using Banker's Algorithm.

C Program:

```
#include <stdio.h>

int main()
{
    int n, m, i, j, k;
    int allocation[10][10], max[10][10];
    int need[10][10];
    int available[10];
    int finish[10] = {0};
    int safeSeq[10];
    int count = 0;

    printf("Enter Number of Processes: ");
    scanf("%d", &n);

    printf("Enter Number of Resources: ");
    scanf("%d", &m);

    printf("\nEnter Allocation Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &allocation[i][j]);

    printf("\nEnter Maximum Matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            scanf("%d", &max[i][j]);

    printf("\nEnter Available Resources:\n");
    for(i = 0; i < m; i++)
        scanf("%d", &available[i]);

    for(i = 0; i < n; i++)
        for(j = 0; j < m; j++)
            need[i][j] = max[i][j] - allocation[i][j];

    while(count < n)
    {
        int found = 0;
        for(i = 0; i < n; i++)
        {
            if(finish[i] == 0)
            {
                for(j = 0; j < m; j++)
                    if(need[i][j] > available[j]) break;

                if(j == m)
                {
                    for(k = 0; k < m; k++)
                        available[k] += allocation[i][k];
                    safeSeq[count++] = i;
                    finish[i] = 1;
                    found = 1;
                }
            }
        }
        if(found == 0) { printf("\nSystem is NOT in Safe State\n"); return 0; }
    }

    printf("\nSystem is in Safe State\n");
    printf("Safe Sequence: ");
    for(i = 0; i < n; i++) printf("P%d ", safeSeq[i]);
    printf("\n");

    return 0;
}
```

Compilation: `gcc bankers.c -o bankers`

Output (5 Processes, 3 Resources, Available: 3 3 2):

```
System is in Safe State
Safe Sequence: P1 P3 P4 P0 P2
```

Shell Script:

```
#!/bin/bash
```

```
echo "Enter Number of Processes:"  
read n
```

```
echo "Enter Safe Sequence (space separated):"  
read -a seq
```

```
echo "Safe Sequence is:"
```

```
for ((i=0;i<n;i++))  
do  
    echo -n "P${seq[$i]} "  
done
```

```
echo  
echo "System is in Safe State"
```

Result: Banker's Algorithm for Deadlock Avoidance was implemented successfully.
